

READING NOTES

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1. INTRODUCTION

The motivation behind this running record of reading notes on mathematics is to gather the ongoing observations, ideas and questions which arise during my study. The entries are primarily for my own understanding, but I hope they may also be of interest to others who enjoy exploring mathematics. As my understanding evolves, some notes may be revised in the future.

2. NOTES

08th Aug, 2026. Mathematics on this day included some readings on the realisation of groups as metric spaces. We start by taking a group G and its generating set S . Now, for any two elements $g_1, g_2 \in G$ we define the *distance* between them to be the word length $g_1^{-1}g_2$ with respect to S . This induces a metric on G , referred to as the *word metric*. Well, to be precise, the word metric is defined as

$$d_S(g_1, g_2) = |g_1^{-1}g_2|_S,$$

where $|g_1^{-1}g_2|_S$ denotes the minimum number of generators from $S \cup S^{-1}$ needed to express $g_1^{-1}g_2$. Moreover, when we think of groups in terms of their Cayley graphs, this word metric is precisely the *path metric* of the graph.

Another interesting thing about viewing groups as metric spaces is the analogue of isomorphisms which we obtain in the latter, called an *isometry*. An isometry is simply a bijective function, say f defined as:

$$f : (X, d_X) \rightarrow (Y, d_Y),$$

such that

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2)$$

for all pairs $x_1, x_2 \in X$.

Another interesting notion is that of the group of symmetries of a metric space X , denoted as $\text{Isom}(X)$. And a group action of a group G on a metric space X would then be the following group homomorphism

$$G \rightarrow \text{Isom}(X).$$

So essentially, for each element $g \in G$, the map gives us an isometry $X \rightarrow X$.

07th Aug, 2026. Some mathematics discussions¹ on this day included an alternate proof of the *Bolzano–Weierstrass theorem*. The idea was to start with a bounded sequence $(a_n) \subseteq [a, b]$, divide the interval into two parts, and keep the part which contains infinitely many terms of the sequence. One can keep doing this, obtaining nested intervals

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

¹Please refer to this [webpage](#) and the notes provided there for these ideas.

whose lengths tend to zero. One can then choose a subsequence by taking a term from each of these intervals, and this subsequence will converge to the unique point determined by the nested intervals. I particularly liked the rather physical analogy used to explain this.

Also, the *Archimedean property* can be realised through the Tortoise and Hare story, and the *Monotone Convergence Theorem* can be visualised by repeatedly partitioning a bounded interval and narrowing down towards a certain point.

In another news, an algebraic set can be defined using an ideal rather than a collection of polynomials. Given an ideal $\mathfrak{J} \subseteq k[x_1, \dots, x_n]$, we can consider

$$V(\mathfrak{J}) = \{a \in \mathbb{A}_k^n : f(a) = 0 \text{ for every } f \in \mathfrak{J}\}$$

Moreover, the radical of an ideal has some interesting properties that one can play around with in commutative algebra. In particular,

$$V(\mathfrak{J}) = V(\sqrt{\mathfrak{J}}),$$

which is quite interesting since the algebraic set cannot distinguish between an ideal and its radical.

Modules, on the other hand, generally have two ways of being viewed. While I prefer the ring-action approach,

$$R \times M \longrightarrow M,$$

I hope to learn to appreciate the linear algebra approach as well, where modules over rings are considered analogous to vector spaces over a field.

05th Aug, 2026. One of the more surprising ideas I encountered today was that the common notion of the zero set of a collection of polynomials is elevated to a central geometric object in algebraic geometry. Such a set is called an *algebraic set*, while a *variety* is, roughly speaking, an irreducible algebraic set (depending on the convention being used).

I also found it interesting that these objects are studied in affine space ($\mathbb{A}_k^n$). Although affine space has points with coordinates such as $(0, 0)$ or, more generally, $(0, \dots, 0)$, it is not regarded as having a distinguished origin in the geometric sense. The point $(0, \dots, 0)$ is then simply another point of the space, and whether it belongs to the zero set of a polynomial depends entirely on the polynomial itself. For instance, if the constant term is nonzero, then $(0, \dots, 0)$ need not be a zero of that polynomial.

Another pleasing observation was that the Cayley graph of the free group F_2 , with respect to its standard free generators, is the infinite 4-regular tree T_4 . Starting from any vertex, the tree branches outward in a remarkably symmetric fashion, and after the initial four edges, there are $4 \cdot 3^{n-1}$ new vertices at distance n from the starting point.

Moreover, algebraic sets are interesting objects. Although they can be defined over any field, many of their fundamental properties are studied over algebraically closed fields.

There is also the *Zariski topology*, which arises naturally on affine spaces and whose closed sets are precisely the algebraic sets, at least for $n = 1$ and $n = 2$.

04th Aug, 2026. Well, the Cayley graph of a free group with respect to a free basis is an infinite regular tree. This graph has some interesting valencies at each point and it is unlike the finite groups' graphs which contain cycles. In other news, $SL_2(\mathbb{Z})$ has some interesting elements which are also its generators. It would be interesting to explore the geometry of its Cayley graph.

03rd Aug, 2026. Today I learned that automorphisms of a group induce automorphisms of its Cayley graph (provided the generating set is preserved). This is interesting because every automorphism of a group is a permutation of its underlying set, so there is a natural embedding $\text{Aut}(G) \hookrightarrow \text{Sym}(G)$. Thus, the same automorphism acts both as a permutation of the group and as a symmetry of its Cayley graph.